Maximum Likelihood Loss Function Cross Entropy Entropy (physics) Statistics (academic discipline) Machine Learning

What are the differences between maximum likelihood and cross entropy as a loss function?

3 Answers

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Answered Jul 5 · Upvoted by Matthew Li, PhD Natural Language Processing & Machine Learning (2017)

I hate to disagree with other answers, but I have to say that in most (if not all) cases, there is no difference, and the other answers seem to miss this.

For instance, in the binary classification case as stated in one of the answers. Let's start by writing out what the binary cross entropy would be for true labels y and predictions (say from a neural network) $p_\theta(y|x)$. I'm assuming we have a 1/0 encoding of the labels, and our model (the neural network) is parameterized by θ .

$$\mathrm{BCE}(y, x, \theta) = -\sum_{i=1}^{n} y_i \log p_\theta(y|x_i) + (1 - y_i)\log(1 - p_\theta(y|x_i))$$

The "pure optimization" view of training would say optimize the above w.r.t. θ, and your done. Now let's take a probabilistic approach to the same problem, i.e., maximize the likelihood of the data under some probabilistic model. The correct likelihood in the case of binary classification is Bernoulli, such that:

$$p(y|\pi) = \prod_{i=1}^{n} \pi_i^{y_i}(1 - \pi_i)^{1-y_i}$$

Now, what we'll do is train a model (say a neural network) to estimate π, and we'll train it to do so based on the inputs. Let's write out the likelihood function for this model:

$$p(y|x, \theta) = \prod_{i=1}^{n} p_\theta(y|x_i)^{y_i}(1 - p_\theta(y|x_i))^{1-y_i}$$

Now, we'd like to maximize the above function w.r.t. θ (hence *maximum likelihood*). Since maximizing products is burdensome, and the log function is monotone, let's maximize the log-likelihood instead and produce the same solution:

$$\mathcal{L}(\theta; x, y) = \sum_{i=1}^{n} y_i \log p_\theta(y|x_i) + (1 - y_i)\log(1 - p_\theta(y|x_i))$$

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objective functions, so there can be no difference between the resulting model or its characteristics.

This of course, can be extended quite simply to the multiclass case using softmax cross-entropy and the so-called multinoulli likelihood, so there is no difference when doing this for multiclass cases as is typical in, say, neural networks.

The difference between MLE and cross-entropy is that MLE represents a structured and principled approach to modeling and training, and binary/softmax cross-entropy simply represent special cases of that applied to problems that people typically care about.

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Sujay Ry, MS from Virginia Tech
Answered May 14

Maximum likelihood is typically used when finding hyper parameters of generative models[Models which requires knowledge of class conditional probability density function]. For example, if you are working with bayesian models where p(y/x, theta)= [p(x/y;theta) * p(y)]/p(x).

Training a generative model involves making the posterior probability (p(y=k / x)) maximum for a given input x and class label k, which is same as maximizing p(x/y;theta). Hence hyperparameter, theta is found by MLE estimate.

For discriminative models such as neural networks, logistic regression , svm's, we don't care about the class distributions. Instead, given a training sample with input x, and label k, a loss is defined and classical optimization such as stochastic gradient descent is applied to directly reduce the loss and find the parameter theta in y= f(x, theta). One such loss function in descrimative analysis is cross entropy loss.

Angli Liu, works at University of Washington
Answered Jul 5

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Suppose your probability model produces N predictions $\mathbf{y_1}, \dots, \mathbf{y_N}$. Each $\mathbf{y_i}$ is a distribution over K classes, $[\hat{y}_i^{1_i}, \dots, \hat{y}_i^{K_i}]$ where $\sum_{j=1}^{K} \hat{y}_i^{j_i} = 1$.

Furthermore, suppose the ground truth of these N instances are $\mathbf{y_1}, \dots, \mathbf{y_N}$. In each vector only one entry $y_i^{j_i}$ is 1, and other entries are 0.

Your negative log likelihood of these N predictions (i.i.d) will be:

$$\begin{aligned} -\log[Pr(\hat{\mathbf{y}}_\mathbf{1}, \dots, \hat{\mathbf{y}}_\mathbf{N})] &= -\sum_{i=1}^{N} \log[Pr(\hat{\mathbf{y}}_\mathbf{i})] \\ &= -\sum_{i=1}^{N} \log \hat{y}_i^{j_i} \\ &= -\sum_{i=1}^{N} \sum_{j_i=1}^{K} y_i^{j_i} \log \hat{y}_i^{j_i} \\ &= \sum_{i=1}^{N} H(\hat{\mathbf{y}}_\mathbf{i}, \mathbf{y_i}) \end{aligned}$$

which is exactly the sum of the cross entropy over all instances. Therefore, for a probability model to be trained on i.i.d training examples, maximize likelihood → minimize negative log likelihood → minimize cross entropy.

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